

File Edit Selection View Go Run ...

AI Lab Programs

AI Lab Programs

> .dist

.gitattributes

AI_Program_1

AI_Program_1(24BECS124).pdf

AI_Program_2

AI_Program_2(24BECS124).pdf

AI_Program_3

AI_Program_3(24BECS124).pdf

AI_Program_4

AI_Program_4(24BECS124).pdf

AI_Program_5

AI_Program_6 U

AI_Program_7 U

AI_Program_8 1, U

AI_Program_9 U

AI_Program_10 1, U

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AI_Program_6 U

AI_Program_7 U

AI_Program_8 1, U

AI_Program_9 U

AI_Program_10 1, U

AI_Program_10 1, U

AI_Program_10 > ...

```
1 import numpy as np
2
3 def euclidean_distance(a, b):
4     return np.sqrt(np.sum((a - b) ** 2))
5
6 class KMeans:
7     def __init__(self, k=2, max_iters=100):
8         self.k = k
9         self.max_iters = max_iters
10
11     def fit(self, X):
12         # Pick random initial centroids from the dataset
13         idx = np.random.choice(X.shape[0], self.k, replace=False)
14         self.centroids = X[idx]
15
16         for _ in range(self.max_iters):
17             # Assign clusters
18             self.labels = np.array([np.argmin([euclidean_distance(x, c) for c in
19 self.centroids]) for x in X])
20
21             # Recompute Centroids
22             new_centroids = np.array([X[self.labels == i].mean(axis=0) if
23 len(X[self.labels == i]) > 0 else self.centroids[i] for i in range(self.k)])
24             if np.all(self.centroids == new_centroids):
```

Problems 2 Output Debug Console Terminal Ports

Code

PS C:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs> python -u "c:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs\AI_Program_10"

Assigned Labels: [0 1 0 0 1 0]

Calculated Silhouette Validation Score: -0.14822328775848606

PS C:\Users\KAVYA\OneDrive\Desktop\AI Lab Programs>

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AI_Program_10 > ...
23 len(X[self.labels == i]) > 0 else self.centroids[i] for i in range(self.k))
24 if np.all(self.centroids == new_centroids):
26 self.centroids = new_centroids
27 return self.labels
28
29 def silhouette_score_simple(X, labels):
30 # Simple evaluation: Mean within-cluster vs closest-cluster distance
31 scores = []
32 for i, x in enumerate(X):
33 c_idx = labels[i]
34 same_cluster = X[labels == c_idx]
35 other_cluster = X[labels != c_idx]
36
37 if len(same_cluster) > 1:
38 a = np.mean([euclidean_distance(x, original) for original in same_cluster
39 if not np.array_equal(x, original)])
40 else:
41 a = 0
42
43 if len(other_cluster) > 0:
44 b = np.min([np.mean([euclidean_distance(x, o) for o in X[labels == c]])
45 for c in set(labels) if c != c_idx])
46 else:
47 b = 0

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Code + - - - | X

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```
39 if not np.array_equal(x, original)]
43 if len(other_cluster) > 0:
44     b = np.min([np.mean([euclidean_distance(x, o) for o in x[labels == c]])
45 for c in set(labels) if c != c_idx])
46 else:
47     b = 0
48
49 if max(a, b) == 0:
50     scores.append(0)
51 else:
52     scores.append((b - a) / max(a, b))
53 return np.mean(scores)
54
55 # Example Usage
56 if __name__ == "__main__":
57     X = np.array([[1, 2], [1, 4], [1, 0], [10, 2], [10, 4], [10, 0]])
58
59     km = KMeans(k=2)
60     labels = km.fit(X)
61
62     score = silhouette_score_simple(X, labels)
63     print("Assigned Labels:", labels)
64     print("Calculated Silhouette Validation Score:", score)
```

Problems 2

Output

Debug Console

Terminal

Ports

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Python

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